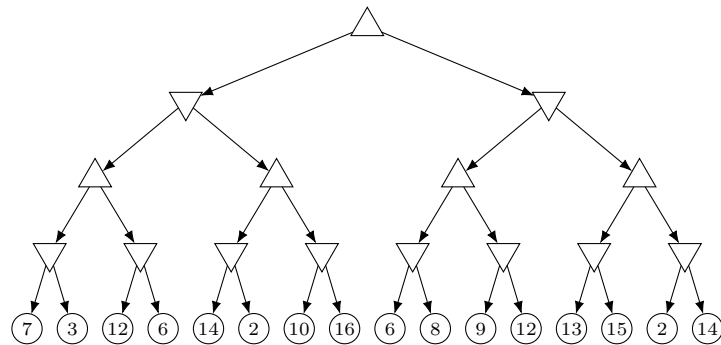


Exercise Sheet 12 — Solutions

Exercise 12.1 (2 marks)

Consider the game inducing the following game tree T . Note that we excluded moves (the edges are unlabeled) because they are irrelevant for the purpose of this exercise.



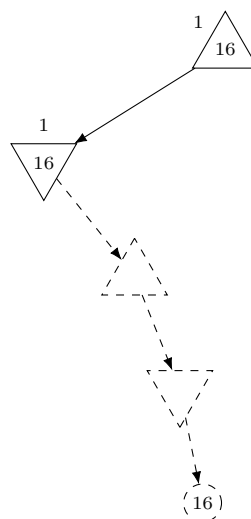
Apply the first 4 iterations of Monte Carlo Tree Search to T with the following policies:

- tree policy: For MAX nodes select the successor with highest utility, for MIN nodes select the successor with lowest utility.
- expansion: Pick the leftmost unexplored child.
- default policy: Pick the rightmost child.

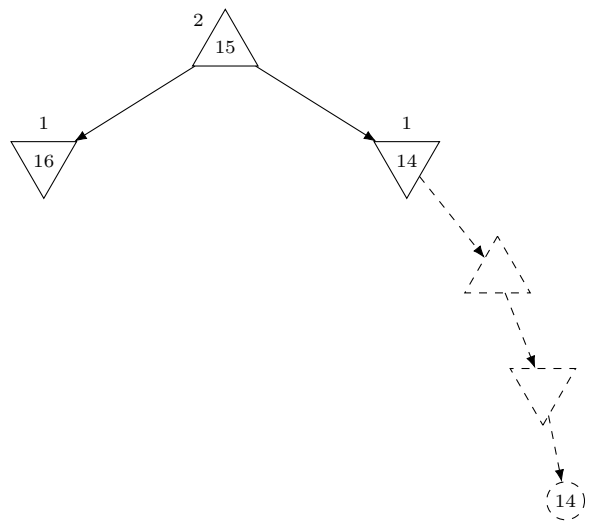
For each iteration, draw the search tree as follows: draw the already explored part of the search tree using solid lines and draw nodes not part of the search tree but encountered during simulation of that iteration using dashed lines. Furthermore, annotate all visited nodes (of the explored part) with the updated utility estimate and visit counter, and annotate the terminal node of the simulation phase with its value.

Solution:

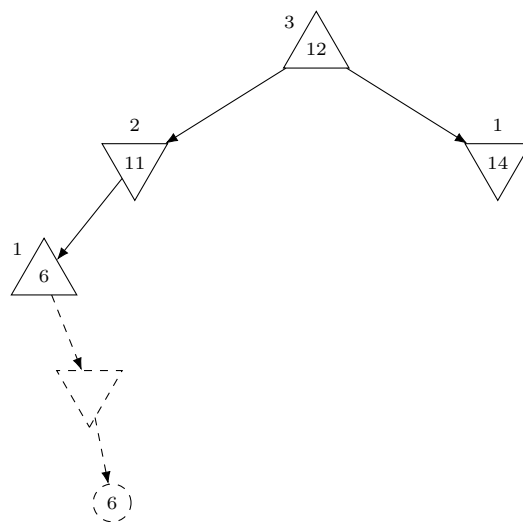
First iteration:



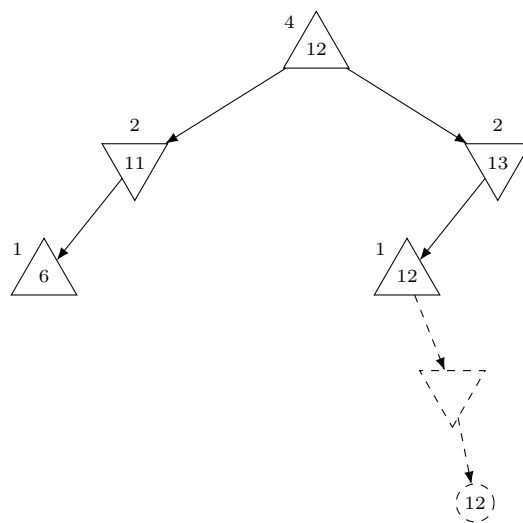
Second iteration:



Third iteration:

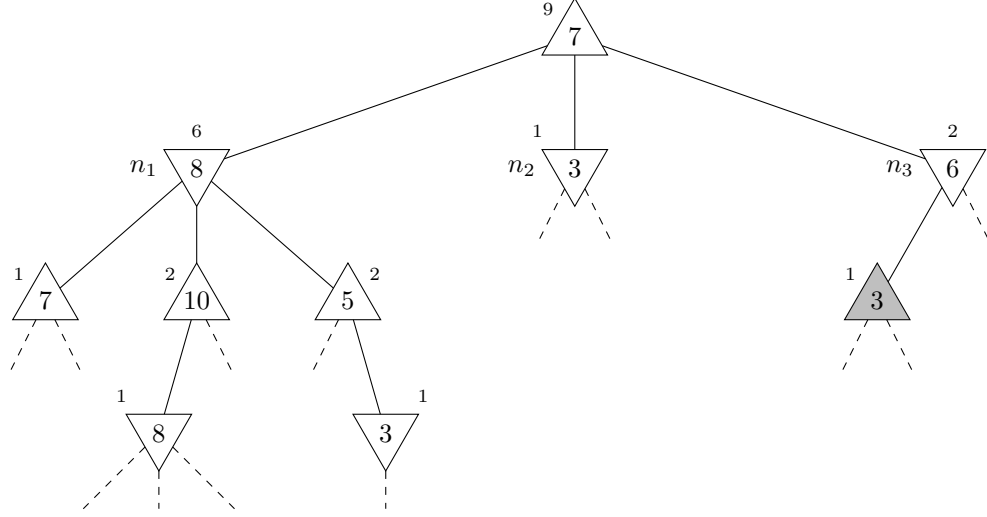


Fourth iteration:



Exercise 12.2 (0.5+0.5+1+1 marks)

Consider the following partial game tree that was created by Monte-Carlo tree search. State value estimates are given inside each search node and visit counters above each search node. Dashed lines indicate missing successors of search nodes.



- Is there a tree policy that guarantees that Monte-Carlo tree search adds one of the children of the gray search node in the next iteration? Justify your answer.
- Compute the probabilities to choose any of the successors n_1 , n_2 or n_3 of the root node using the ϵ -greedy tree policy with $\epsilon = 0.2$.
- Compute the probabilities to choose any of the successors n_1 , n_2 or n_3 of the root node using the softmax tree policy with $\tau = 10$. Provide how you derived your solution.
- Compute the probabilities to choose any of the successors n_1 , n_2 or n_3 of the root node using the UCB1 tree policy. Provide how you derived your solution.

Solution:

- No. The parent of the gray node still has a successor that is not part of the tree, which must be added before a successor of the gray node can be added.
- The probabilities are $\mathbb{P}(n_1) = 0.8$, $\mathbb{P}(n_2) = 0.1$ and $\mathbb{P}(n_3) = 0.1$.
- We know that $\mathbb{P}(n)$ is proportional to $e^{\frac{\hat{v}}{\tau}}$, so we first compute this value for the three nodes: $b_1 = e^{\frac{8}{10}} \approx 2.226$, $b_2 = e^{\frac{3}{10}} \approx 1.350$ and $b_3 = e^{\frac{6}{10}} \approx 1.822$. We hence get the probabilities $\mathbb{P}(n_1) = \frac{b_1}{b_1+b_2+b_3} \approx 0.41$, $\mathbb{P}(n_2) = \frac{b_2}{b_1+b_2+b_3} \approx 0.25$ and $\mathbb{P}(n_3) = \frac{b_3}{b_1+b_2+b_3} \approx 0.34$.
- This is a bit of a trick question. UCB1 selects the state that maximizes $\hat{v} + \sqrt{\frac{2 \cdot \ln P}{N}}$ where N and P are the numbers of samples for the node under consideration and its parent, respectively. So unless multiple states have the same value, UCB1 selects one state with probability 1.

We first compute the bonus terms for all nodes: $B(n_1) = \sqrt{\frac{2 \cdot \ln 9}{6}} \approx 0.856$, $B(n_2) = \sqrt{\frac{2 \cdot \ln 9}{1}} \approx 2.096$ and $B(n_3) = \sqrt{\frac{2 \cdot \ln 9}{2}} \approx 1.482$. Adding the state value estimates so far to these bonus terms leaves us with $\hat{v}_1 \approx 8.856$, $\hat{v}_2 \approx 5.096$ and $\hat{v}_3 \approx 7.482$, so we get $\mathbb{P}(n_1) = 1$ and $\mathbb{P}(n_2) = \mathbb{P}(n_3) = 0$.